

Raju Bandam

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Professional Summary

Results-driven Full-Stack Software Engineer with **5+ years of industry experience** and a Master's in Computer Science (Bowling Green State University, May 2026) specializing in designing and delivering **high-throughput, fault-tolerant distributed systems** and **modern Angular-based frontend applications** used by thousands of concurrent users. Proven track record of building production-grade solutions using **Java, Spring Boot, Angular 18+, Microservices, REST APIs, and AWS** that drive measurable business outcomes — including **30–45% performance improvements**, reduced deployment cycles, and improved system reliability.

Seeking a backend, full-stack, or software engineering role where I can architect scalable systems, build intuitive user interfaces, mentor peers, and contribute to engineering excellence at a high-growth technology organization. Known for writing clean, testable, well-documented code and translating complex technical requirements into reliable, maintainable solutions delivered on time.

Technical Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C++

Frontend: Angular 18+, Angular Signals, RxJS, NgRx, Angular Material, Standalone Components, Lazy Loading, PrimeNG

Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, REST APIs, GraphQL, Microservices, OAuth2, JWT

Cloud / DevOps: AWS (EC2, S3, IAM, RDS, Lambda), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines

Databases: MySQL, PostgreSQL, MongoDB, Redis (Caching)

Messaging / Async: Apache Kafka, RabbitMQ

Testing & Quality: JUnit 5, Mockito, Jasmine, Karma, Integration Testing, SonarQube

Tools & Methodologies: Git, Maven, Postman, JIRA, Agile/Scrum, OOP, SOLID Principles, Design Patterns

Education

Bowling Green State University, Bowling Green, Ohio
Master of Science in Computer Science — GPA: 3.8/4.0

Jul 2024 – May 2026

BV Raju Institute of Technology, Hyderabad, India
Bachelor of Technology in Computer Science

2017 – 2020

Professional Experience

Full-Stack Software Engineer — Cigniti Technologies
Hyderabad, India

Nov 2020 – Oct 2023

- Architected and delivered scalable backend systems using **Java 11** and **Spring Boot**, supporting **10,000+ concurrent users** with 99.9% uptime in production environments.
- Built and maintained **Angular 18+ frontend applications** using standalone components, Angular Signals for reactive state management, and lazy-loaded feature modules – improving initial page load time by **35%**.
- Engineered **30+ RESTful APIs** with JWT/OAuth2 authentication, role-based authorization, request validation, and centralized exception handling, reducing API error rates by **25%**.
- Developed reusable **Angular component libraries** with Angular Material and PrimeNG, standardizing UI patterns across 3 enterprise applications and cutting frontend development time by **30%**.
- Integrated Angular frontend with Spring Boot microservices using **RxJS observables** and **HttpClient interceptors** for centralized token injection, error handling, and retry logic across all API calls.
- Implemented **NgRx state management** for complex enterprise workflows, eliminating redundant API calls and improving UI responsiveness by **40%** through efficient client-side caching.
- Designed and deployed a **microservices architecture** across 8+ independently deployable services, improving release frequency by **40%** and reducing inter-service coupling.
- Optimized complex SQL queries, indexing strategies, and database transactions in **PostgreSQL** and **MySQL**, achieving **30% reduction** in average query execution time.
- Implemented **Redis caching** layer for high-frequency read endpoints, reducing average API latency by **45%** and cutting database load significantly.

- Built and maintained **CI/CD pipelines** using Jenkins and GitHub Actions, reducing deployment time from hours to under **15 minutes** with automated Angular build optimization and zero-downtime releases.
- Containerized **12+ microservices** using Docker and orchestrated deployments via Kubernetes, ensuring consistent environments across dev, staging, and production.
- Deployed and managed cloud infrastructure on **AWS EC2, S3, RDS, and IAM**, supporting auto-scaling configurations that reduced infrastructure costs by **20%**.
- Integrated **Apache Kafka** for asynchronous event-driven communication between services, improving system decoupling and processing throughput for high-volume data pipelines.
- Led root-cause analysis for production incidents, reducing mean time to resolution (**MTTR**) by 35% through structured post-mortems and proactive monitoring improvements.
- Authored **unit and integration tests** using JUnit 5, Mockito, Jasmine, and Karma, achieving **85%+ test coverage** across backend and frontend codebases, reducing post-release defects by 30%.
- Mentored 2 junior engineers and led code reviews across a **6-member Agile team**, improving code quality and accelerating onboarding by 50%.

Projects

iNSta – Intelligent Test Automation Platform *Angular 18, Spring Boot, Flask, PostgreSQL, MongoDB, Redis, Kafka, Docker, AWS*

- Architected a **production-grade, full-stack automation platform** with an Angular 18 frontend and Spring Boot/Flask backend, orchestrating end-to-end QA workflows and reducing manual testing effort by **60%** across 5+ cross-functional teams.
- Built an **Angular 18 single-page application (SPA)** using standalone components, Angular Signals for fine-grained reactivity, and feature-based lazy loading – achieving a **Lighthouse performance score of 90+** and sub-2-second initial load time.
- Designed and implemented **20+ REST APIs** for test scheduling, real-time execution control, reporting pipelines, and analytics dashboards – all secured with JWT-based RBAC supporting **multi-tenant access** across Admin, QA Lead, and Tester roles.
- Developed a **real-time test execution dashboard** in Angular using WebSocket integration and RxJS streams, providing live status updates, progress indicators, and execution logs without page refresh.
- Built a **dynamic form system** using Angular Reactive Forms with runtime validation rules loaded from the backend API, supporting configurable test parameter inputs across 50+ test suite types.
- Implemented **NgRx store** for global application state covering active test runs, user session, and filter preferences – reducing redundant API calls by **45%** and enabling seamless navigation between modules.
- Engineered a **hybrid data persistence layer** combining PostgreSQL for structured test metadata and MongoDB for unstructured logs and execution artifacts, optimized to support **1M+ test records** without degradation.
- Implemented an **asynchronous job processing engine** using Apache Kafka and thread pool executors, eliminating request blocking and improving overall system throughput by **50%**.
- Reduced API response times by **40%** through Redis caching, connection pooling, and targeted PostgreSQL index tuning on high-frequency query paths.
- Containerized all services using **Docker** and automated deployment pipelines via Jenkins for consistent promotion across dev, staging, and production.
- Established a **centralized structured logging system** with correlation IDs per test run, reducing average debugging time by **35%** across distributed services.
- Achieved **90%+ test coverage** using JUnit 5, Mockito (backend) and Jasmine/Karma (Angular frontend), with SonarQube quality gates enforced in the CI pipeline.
- Delivered **API documentation** using Swagger/OpenAPI 3.0 and Angular Storybook for UI component documentation, reducing integration time for consumer teams by **40%**.

Food Delivery Application *Angular 18, Spring Boot, MySQL, Redis, AWS EC2, Docker, Kafka, JWT*

- Designed and built a **full-stack food delivery platform** with an Angular 18 frontend and Spring Boot backend, covering restaurant discovery, cart management, order processing, payment, and real-time delivery tracking – handling **5,000+ transactions/day** at peak load.
- Developed a **responsive Angular 18 SPA** with standalone components and Angular Material, featuring dynamic restaurant listings, real-time cart updates with Signals, and an order tracking view – optimized for both desktop and mobile viewports.
- Implemented **Angular route guards and lazy-loaded modules** for Customer, Restaurant Owner, and Delivery Agent portals, reducing bundle size by **40%** and improving role-based navigation security.
- Built a **real-time order tracking UI** using Angular with WebSocket/Kafka consumer integration, displaying live delivery agent location updates and order status transitions with zero polling overhead.
- Developed a **secure user authentication flow** in Angular using JWT token management with HttpClient interceptors for automatic token attachment, silent refresh, and redirect-on-expiry behavior.

- Architected a **multi-state order lifecycle engine** with strict state machine transitions (Placed → Confirmed → Preparing → Out for Delivery → Delivered / Cancelled), backed by audit logging for every state change.
- Integrated a **third-party payment gateway** (Stripe-compatible) with idempotent request handling and webhook-based payment confirmation, achieving a **99.8% payment success rate** with zero duplicate charge incidents.
- Built a **Kafka-based real-time event streaming pipeline** propagating order state changes across Order, Notification, Delivery, and Analytics services with **sub-second latency** and guaranteed message delivery.
- Implemented a **geo-proximity delivery matching algorithm** to assign the nearest available delivery agent dynamically, reducing average assignment time by **30%**.
- Reduced API latency by **35%** through Redis caching of menus, restaurant listings, and user sessions, combined with composite index strategies on MySQL.
- Deployed on **AWS EC2 with Auto Scaling Groups and Application Load Balancer (ALB)**, maintaining **99.95% monthly uptime** during traffic spikes.
- Wrote **comprehensive test suites** with JUnit 5 and Mockito (backend) and Jasmine/Karma (Angular), maintaining **85%+ code coverage** with Jenkins CI quality gates.

Certifications & Achievements

- AWS Certified Developer – Associate (*Azure Developer Associate, DevOps Engineer Expert*)
- Completed **500+ LeetCode problems** (Medium/Hard) focusing on graphs, dynamic programming, and system design
- Academic Excellence Award – BGSU Computer Science Department, 2025